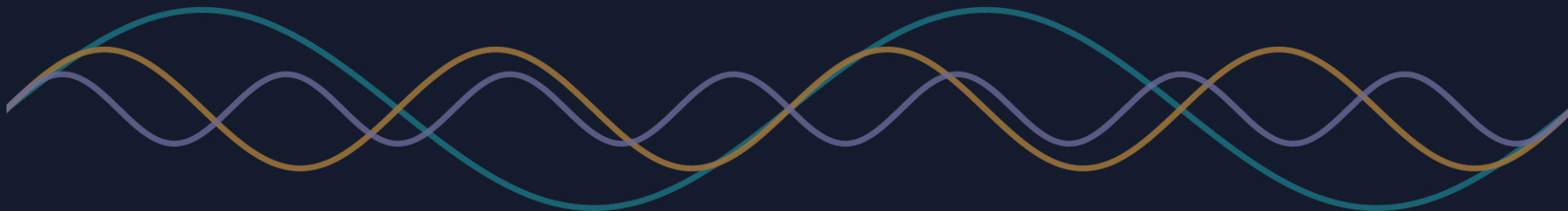


Discovering the Fourier Basis and the Multiplication Law

— *from convolution data alone*

Given only random circular-convolution samples (x, h, y) , can a model recover BOTH the DFT and pointwise complex multiplication — without either being written into the model?



One question, two hidden objects

If the only data is $\mathbf{y} = \mathbf{x} \circledast \mathbf{h}$ for many random $\mathbf{x}, \mathbf{h} \dots$
can a model learn the structure that makes convolution simple?
convolution in the time domain is equivalent to pointwise multiplication in the frequency domain

1

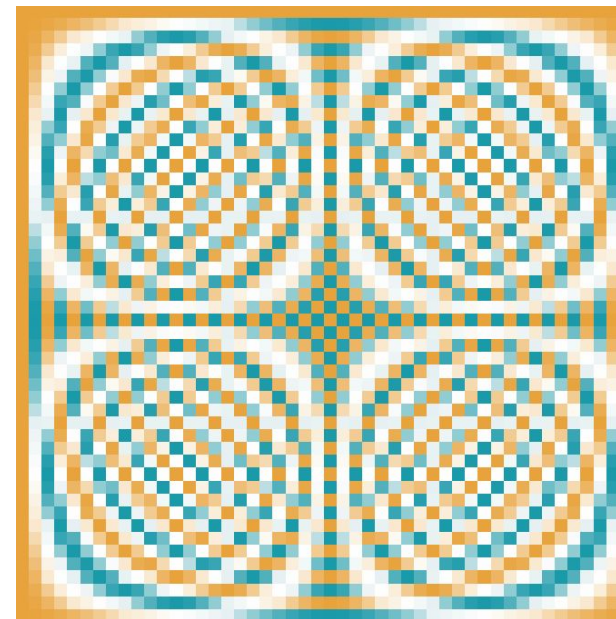
A linear basis that decouples convolution

→ *DFT*

2

The per-coordinate interaction law

→ $\times$
multiply



*Re(DFT) — the hidden structure the
model is asked to rediscover*

Why the convolution theorem matters



Signal processing

Filtering = convolution. The Fourier transform turns time-domain convolution into frequency-domain pointwise multiplication.



Time series

Smoothing, periodicity, seasonality, and long-range dependence are all 1-D convolutions — and FFT accelerates them.



Computer vision

CNNs are stacks of convolutions. 1-D circular convolution is the cleanest, most verifiable version of that structure.

A small AI-for-math experiment: give the data and one weak structural hint, withhold the DFT and multiplication, and see whether the classical theorem re-emerges from the numbers.

Circular convolution is a circulant matrix

Definition (length- n , indices mod n):

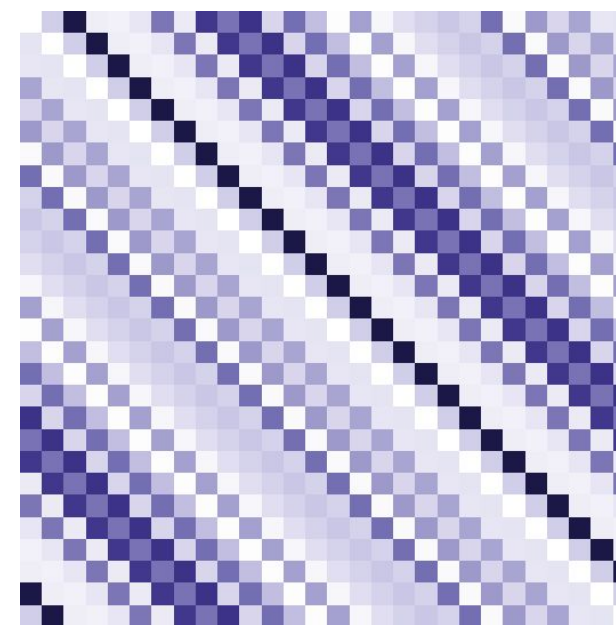
$$y[t] = \sum_s x[s] \cdot h[(t - s) \bmod n]$$

Implemented directly — no FFT, no built-in convolution.

Fix h : the map $x \mapsto x \circledast h$ is linear, so

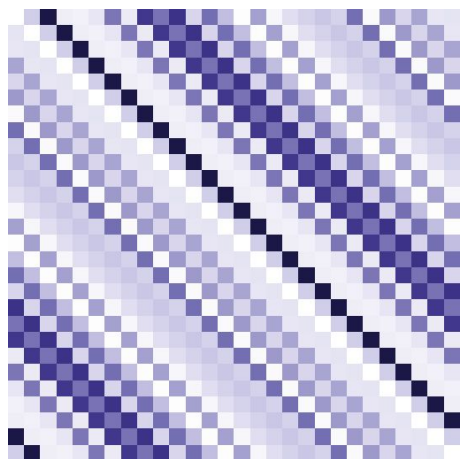
$$\mathbf{y} = \mathbf{C} \mathbf{x} \quad \text{where } \mathbf{C} \text{ is the circulant matrix generated by } h$$

(each row is the previous row shifted by one).



$|\mathbf{C}|$ — the shifted-band structure of a circulant

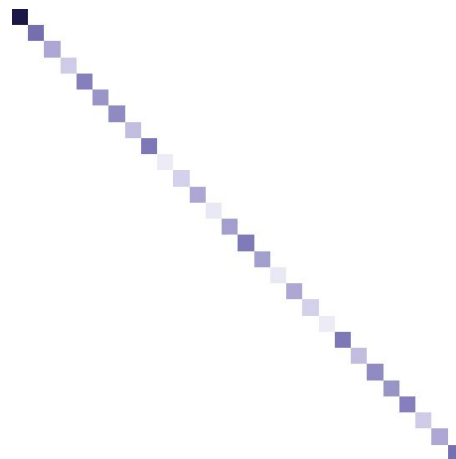
The DFT diagonalizes every circulant



C (dense)



$$F \left(\cdot \right) F^{-1}$$



$F C F^{-1}$ (diagonal)

Convolution theorem

$$F(\mathbf{x} \circledast \mathbf{h}) = (F\mathbf{x}) \odot (F\mathbf{h})$$

equivalently

$$\mathbf{x} \circledast \mathbf{h} = F^{-1}((F\mathbf{x}) \odot (F\mathbf{h}))$$

$$F[j, k] = e^{(-2\pi i jk/n)}$$

Key fact: all circulant matrices share one eigenbasis — the DFT. That shared basis is exactly what the model must rediscover.

We never write the answer into the model

Hard constraints (main model)

-  No FFT in data, convolution, model, or training code
-  No built-in convolution — circular conv by definition
-  No hard-coded elementwise multiplication in models.py
-  A is an unconstrained complex matrix (not DFT/unitary/circulant)
-  DFT lives only in analysis / baselines, for evaluation



The ONE assumption we keep

There exists a linear basis in which each output coordinate depends only on the same-index input pair — through a single shared function.

Not the DFT. Not multiplication. Just: “some basis makes the coordinates decouple.”

Random triples (x, h, y)

Generation

$$\mathbf{x} \sim \mathcal{N}(0, \mathbf{I})$$

$$\mathbf{h} \sim \mathcal{N}(0, \mathbf{I})$$

$$\mathbf{y} = \mathbf{x} \circledast \mathbf{h} \quad (\text{hand-written circular convolution})$$



Why Gaussian?

Random Gaussian vectors cover the whole space, so the model must learn the convolution operator itself — not memorize a few directions.

Out-of-distribution kernels (test only)



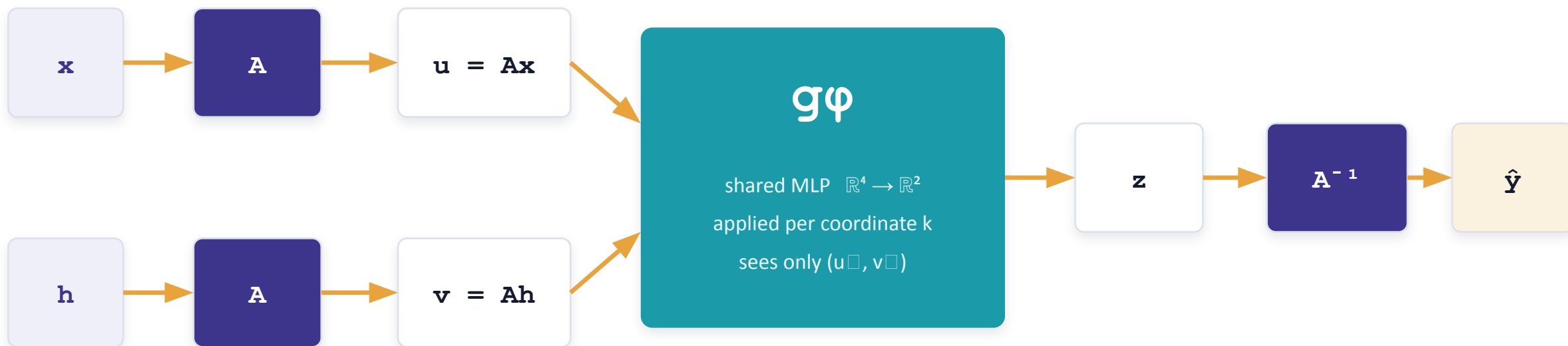
Sparse kernel — only a few non-zero positions



Structured kernel — noise $\times$ a decaying envelope

Test: if the model truly learned the convolution algebra, a new kernel distribution should still work.

Learned basis + shared coordinate-wise MLP



Nothing in this pipeline computes $\mathbf{u} \cdot \mathbf{v}$. $\mathbf{A}$ is a free complex matrix; $g\phi$ is a generic MLP. Both the basis and the multiplication must be learned.

Assumed vs. discovered



DISCOVERED

Change of basis A

A free matrix is given a place — but which basis it becomes is learned.

→ *the DFT (up to gauge)*



ASSUMED

Coordinate decoupling

Each output uses only the same-index pair, via one shared $g\phi$.

the single weak prior



DISCOVERED

The per-coordinate operation

$g\phi$ is free — it could learn add, max, anything.

→ *complex multiplication*

Only the decoupling is assumed. The basis and the law are forced to emerge — because coordinate-wise-in-some-basis can only mean the DFT, and multiplication.

Stabilize the basis, then learn the law

1 Basis discovery

Train A with output loss + a diagonalization loss $L_{\text{diag}} = \text{off-diagonal energy of } A C \square A^{-1}$. Not DFT supervision — it only says “decouple the coordinates.”



2 Freeze A , learn $g\phi$

With the coordinate system fixed, the shared MLP learns the per-coordinate law without chasing a moving basis.



3 Fine-tune $g\phi$

A low-learning-rate polish of the interaction law.

Four readouts — MSE alone is not enough



Test MSE

Is the predicted convolution accurate?
 $\text{mean } |\hat{y} - y|^2$



Diagonalization residual

Off-diagonal energy of $A C A^{-1}$. $\rightarrow 0$ means A decouples convolution.



DFT row residual

Best-match each row of A to a DFT row (allow permutation + phase + scale).



Multiplication probe

Compare $g\phi(u,v)$ to $c \cdot u \cdot v$. Small residual = the law is multiplication.

The multiplication probe is the decisive new readout: it inspects the learned function directly, not just the fit.

It scales: $n = 16, 32, 64$

n	Test MSE	Diag. residual	DFT row residual	Mult. probe
16	$1.61e-3$	$1.55e-6$	$8.42e-4$	$2.39e-3$
32	$1.02e-3$	$1.07e-5$	$2.25e-3$	$3.38e-3$
64	$1.09e-3$	$8.16e-6$	$2.00e-3$	$1.54e-3$

OOD generalization

Unseen sparse & structured kernels:

$1e-4 - 1e-5$

MSE — the operator transfers, not just the training distribution.



Test MSE $\sim 1e-3$ at every size



A rows align tightly with the DFT

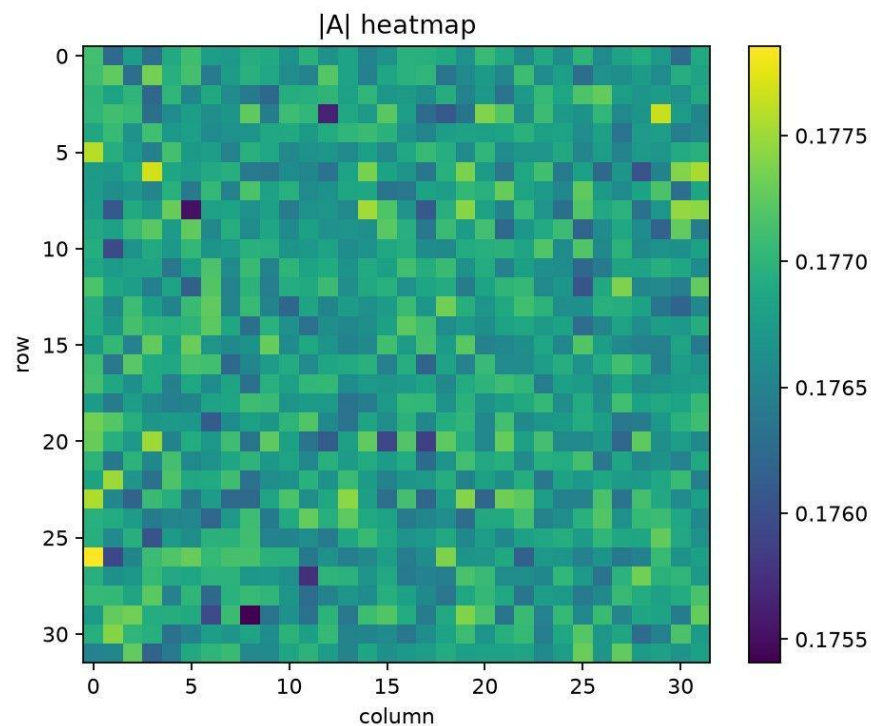


Diagonalization residual ≈ 0 — A decouples convolution

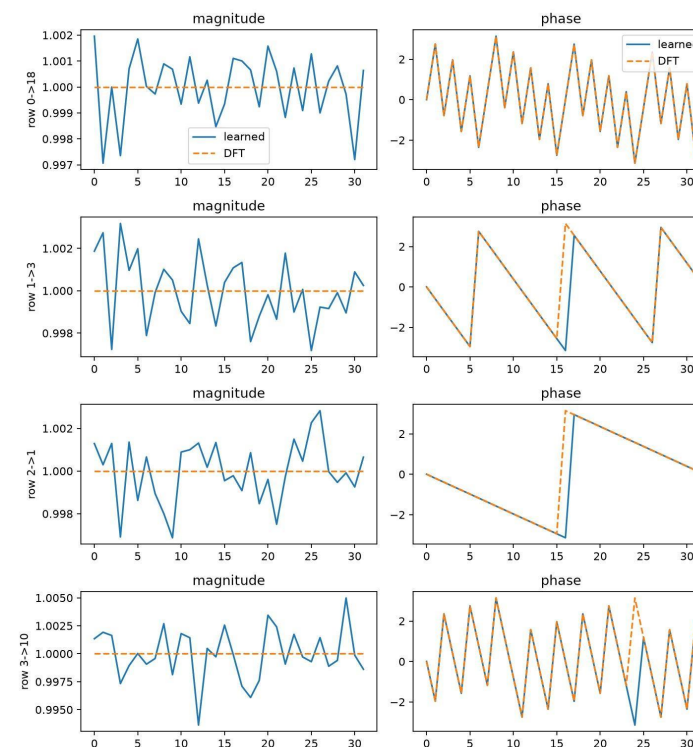


Probe $1.5e-3$ – $3.4e-3$ — $g\phi \approx$ multiplication

The learned basis is the DFT

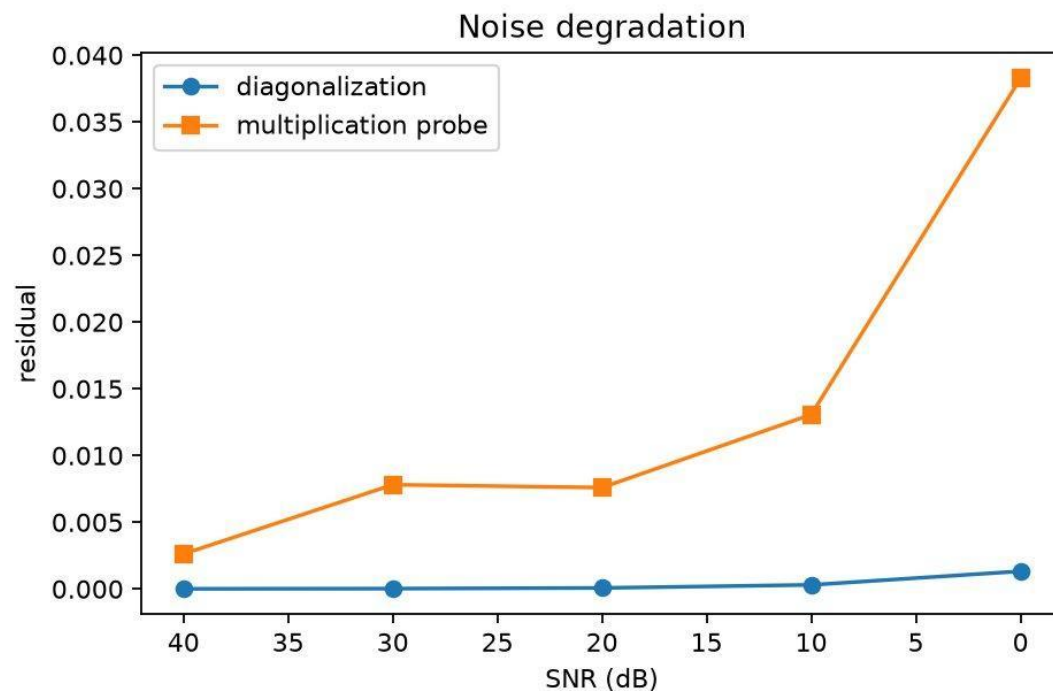


$|A| \approx 1/\sqrt{32} \approx 0.177$ — uniform magnitude on every entry, just like the DFT. The gauge fix removed all row-scale drift.



Rows match the DFT — learned (blue) vs DFT (orange): magnitudes ≈ 1 and phases coincide. The $\pm\pi$ phase jumps are wrap-around, not error.

Graceful degradation under label noise



What the curves say

At 40 dB basis and law are recovered cleanly (residuals $1e-6 - 1e-3$).

As SNR ↓ both residuals grow — the multiplication-probe residual climbs steeply, while diagonalization stays low. The law degrades faster than the basis.

Cleaner data → a sharper Fourier basis and a cleaner multiplication law. Exactly the expected trend.

Baselines & controls

Model	Test MSE	Note / params
SpectralNeuralModel (ours, n=32)	1.02e-3	learned A + MLP — NO hard product
HardProductOracle	5.48e-10	true DFT + product (reference)
HardProductLearnedBasis	1.03e-11	learned A + product (reference)
BilinearTensor	3.44e-2	32,768 params
MLPControl	2.49e-1	156,448 params
NonlinearEncoderControl	8.25e0	34,754 params



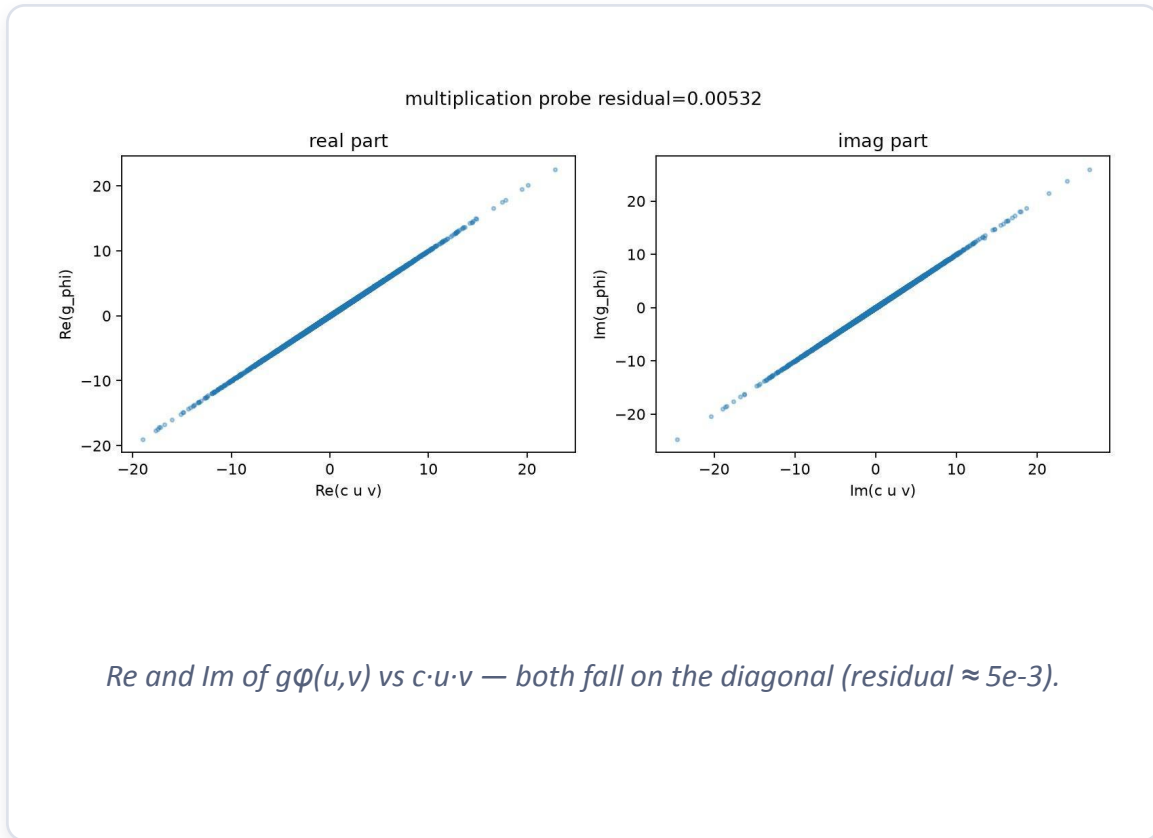
Why this matters

Our model has no hard-coded product, yet reaches ~1e-3.

A generic MLP with 150k+ parameters is two orders of magnitude worse.

The coordinate-independent linear basis — not raw capacity — is what recovers the convolution algebra.

The multiplication law emerged



$$g\phi(u, v) \approx c \cdot u \cdot v$$

Probe residual **1.5e-3 – 3.4e-3** across $n = 16, 32, 64$

- ✓ The network was never told to multiply — only to act coordinate-wise.
- ✓ Fitting one global complex scale c , $g\phi$ collapses onto the exact product.
- ✓ Holds on held-out (u, v) from fresh x, h — a law, not memorization.

CONCLUSION

What we found

Discovered from data

- ✓ $A \approx \text{DFT}$, up to permutation / phase / scale
- ✓ A diagonalizes every circulant operator
- ✓ A 's rows align tightly with the DFT
- ✓ $g\phi(u,v) \approx c \cdot u \cdot v$ — multiplication
- ✓ Consistent across $n = 16, 32, 64$

learned linear basis + shared coordinate-wise MLP $\Rightarrow$ *the Fourier basis and the multiplication law grow out of the data.*